Imagine, Interpret, Act: Giving Temporal Latents a Sense of Progress

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Abstract

Predicting how an action changes the world is not the same as understanding whether that change advances a task. Biological control motivates a lightweight internal-world-model view: anticipate an action's consequence, then interpret what that consequence means for the current goal. We ask whether predictive visual latents can acquire this sense of progress. LeWM offers a clue: its representation retains physical state information while consecutive latent changes become increasingly aligned during training. This emergent straightening, however, has no goal semantics; a straight trajectory may advance, stall, or move away from a selected goal. **We introduce Latent Sense of Progress (LSP), comprising LSP shaping and LSP search.** LSP shaping uses a lightweight Bellman-style objective over ordered trajectories to organize the shared predictive latent into a goal-relative progress coordinate; it requires no reward, success, or phase labels and no separate progress network. On held-out Cube trajectories, LSP shaping yields higher straightness, goal-order accuracy, and within-trajectory rank correlation, with only a small change in next-latent prediction error. LSP search then queries progress once per replan to initialize stage-compatible action proposals and refines global elites with a balanced multi-center update; the original world-model terminal cost remains the sole evaluator. Across four visual-control tasks with matched rollout budgets, LeWM + LSP improves closed-loop success over LeWM + CEM by 4.7–14.7 percentage points, with 95% intervals excluding zero on three.

Introduction

Predictive latents can describe how the world changes without indicating whether that change advances a chosen goal. They may encode pose, contact, and action-conditioned dynamics, while latent goal planners typically introduce the goal only through an endpoint comparison after rollout (Zhou et al. 2025; Maes et al. 2026). This comparison makes the goal usable as a cost but leaves progress implicit in the representation. Consequently, the same physical transition may advance one goal, be irrelevant to another, or move away from a third. The representational challenge is to express this goal-relative meaning without discarding the physical information required for prediction.

Biological motor control separates predicting consequences from interpreting them for action. Internal forward models predict the next motor state from the current state and motor command (Wolpert, Ghahramani, and Jordan 1995; Miall and Wolpert 1996), while model-based choice relates predicted transitions to outcomes and premotor activity can maintain several potential reaches before selection (Gläscher et al. 2010; Cisek and Kalaska 2002, 2005). These findings motivate an imagine–interpret–act abstraction: predict consequences in an internal state, interpret them relative to the goal, and retain alternatives until selection. Predictive temporal latents provide a natural substrate for this abstraction. LeWM reports that its shared latent retains physical state information and that consecutive latent displacements become increasingly aligned during predictive training, despite no straightening loss (Maes et al. 2026); explicit curvature regularization can similarly improve latent-planning geometry (Wang et al. 2026). Yet straightness is invariant to temporal orientation and goal identity: it cannot distinguish advance, stall, or reversal relative to a selected goal. The challenge is to turn temporal organization into goal-relative progress without erasing predictive content.

We introduce Latent Sense of Progress (LSP) to meet this challenge through two components. LSP shaping organizes the shared predictive latent into a goal-relative coordinate, and LSP search uses that coordinate to guide action search. A cosine query reads local progress toward a selected future observation. LSP shaping uses a low-weight trajectory-level Bellman objective, constructed from temporally ordered action windows, to organize the latent around this query without reward, success, or phase labels and without a separate progress head (Ma et al. 2023). The resulting scalar is goal relative rather than an absolute completion percentage or a phase class. Because progress is read from the full latent, LSP shaping remains coupled to next-latent prediction and must preserve predictive content.

To determine whether this coordinate is actionable, LSP search exposes it through a deliberately narrow planning interface (Figure 1). Each replan queries progress once to construct stage-compatible initial proposals, then partitions the global elites into balanced centers for local refinement. The original terminal latent cost alone ranks candidates and se-

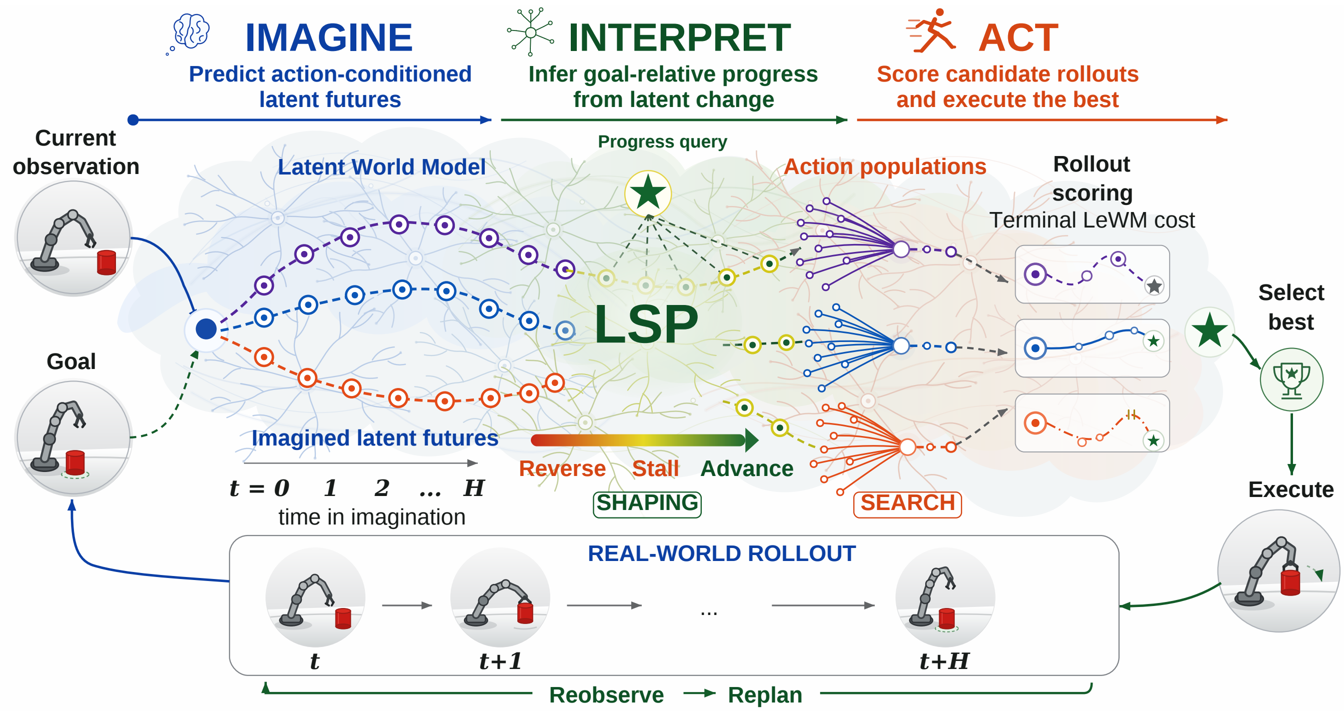

Figure 1: Conceptual overview of LSP. SHAPING organizes predictive latents into a goal-relative progress coordinate; SEARCH conditions initial action proposals on this coordinate and splits global elites into balanced centers for refinement. The terminal world-model cost remains the sole evaluator.

lects the final center. Thus LSP search changes how search is initialized and how elite modes are refined without replacing the world model's evaluation rule.

We evaluate the representation and planning claims in sequence. On held-out Cube trajectories, LSP shaping raises straightness from 0.319 to 0.366, goal-order accuracy from 82.1% to 83.5%, and within-trajectory rank correlation from 0.665 to 0.693, while next-latent MSE rises by only 2.1%. Proposal diagnostics then test whether the learned coordinate structures the first action population, and paired evaluations on four tasks measure downstream utility under matched rollout budgets. LeWM + LSP improves closed-loop success by 4.7–14.7 percentage points, with 95% intervals excluding zero on three of four tasks. Together, these experiments support three contributions:

- **From temporal straightness to goal-relative progress.** We identify temporal orientation and goal identity as the missing semantics of straight predictive trajectories, then introduce a trajectory-only Bellman-style objective that embeds progress in LeWM's shared latent without outcome or phase labels or a separate progress network.
- **A minimal progress-to-planning interface.** We query progress once to condition stage-compatible action proposals and retain alternative elite modes through balanced refinement, while leaving the learned dynamics and terminal evaluator unchanged.
- **Evidence from representation to planning.** Paired latent analyses test whether LeWM's temporal organization can become goal-relative progress; proposal and control analyses then show how this coordinate changes the first action population and closed-loop planning. LeWM + LSP improves success by 4.7–14.7 percentage points across four tasks, with intervals excluding zero on three.

Related Work

Latent World Models for Visual Planning. Latent world models plan by rolling action-conditioned representations forward and comparing a predicted terminal latent with a goal. DINO-WM uses pretrained visual features, whereas PLDM and LeWM learn reward-free joint-embedding predictors from offline pixels (Zhou et al. 2025; Sobal et al. 2025; Maes et al. 2026). LeWM also retains physical state and develops alignment between consecutive latent displacements. These models predict change, but their endpoint costs leave intermediate progress implicit; LSP exposes goal-relative progress before action search.

Biological Motivation for Goal-Relative Progress. Internal forward models predict the sensory consequences of motor commands (Wolpert, Ghahramani, and Jordan 1995; Miall and Wolpert 1996), while model-based choice relates transitions to outcomes and premotor activity can maintain several potential reaches (Gläscher et al. 2010; Cisek and Kalaska 2002, 2005). This motivates separating consequence prediction, goal-relative interpretation, and action selection.

From Temporal Geometry to Progress. Temporal Straightening explicitly reduces curvature, whereas LeWM observes straightening as an emergent property (Wang et al. 2026; Maes et al. 2026); neither specifies temporal direction or goal identity. VIP learns value-like visual representations, HILP learns policy directions, FOUNDER uses temporal distance as an imagination reward, and SCoTS uses distance-preserving latents for trajectory stitching (Ma et al. 2023; Park, Kreiman, and Levine 2024; Wang et al. 2025; Lee and Choi 2025). LSP instead embeds goal-relative progress in the full predictive latent without outcome labels or a separate progress network.

Proposal-Guided, Mode-Preserving Search. Conditional generative models represent multiple goal-compatible sub-trajectories, and TD-MPC2 mixes learned policy trajectories with sampling-based planning (Kim et al. 2024; Hansen, Su, and Wang 2024). CEM refits low-cost samples; iCEM improves reuse and noise scheduling, Lazy Cross-Entropy Search changes continuous-action sampling, and DecentCEM maintains independent populations (Rubinstein and Kroese 2004; Pinneri et al. 2021; Hoerger et al. 2024; Zhang et al. 2022). LSP conditions initial proposals and balances global elites under the unchanged terminal cost; fixed-budget iCEM and DecentCEM control for gains from a stronger optimizer alone.

Method

Problem Setup and Design Overview

We describe LSP for a goal-conditioned latent world model and instantiate it with LeWM in our experiments. Let o_t be the current image, o_g a reachable goal image, and $A = (a_t, \ldots, a_{t+H-1})$ a sequence of macro actions. An encoder E_θ produces $z_t = E_\theta(o_t)$ and $z_g = E_\theta(o_g)$, while an action-conditioned predictor rolls A forward to $\hat{z}_{t+H}(A) = F_\theta(z_t, A)$. Planning minimizes the terminal latent cost

$$C_\theta(A; z_t, z_g) = \|\hat{z}_{t+H}(A) - z_g\|_2^2 . \tag{1}$$

For our LeWM instantiation, C_θ is exactly its released terminal-MSE cost form. LSP shaping changes the checkpoint and hence the learned cost landscape, but not Equation 1.

Standard CEM samples N sequences from one diagonal Gaussian, retains the K_e lowest-cost sequences, and refits one set of moments. This exposes two distinct failures. A *coverage failure* occurs when the initial population contains no useful action sequence. A *collapse failure* occurs when useful sequences are selected but their global moment fit averages away the structure that made them useful. We call the representation objective *LSP shaping*, and the progress-conditioned proposal density plus balanced multi-center CEM *LSP search*; *LeWM + LSP* combines both. LSP shaping organizes temporal latent structure into goal-relative progress, while LSP search uses that progress once to seed and preserve alternative action modes. Progress changes where search begins; Equation 1 alone ranks candidates and selects centers.

LSP Shaping

Temporal straightening is the motivating structure, not the target optimized by LSP shaping. Evidence from predictive latent models suggests that their trajectories can already possess local temporal organization; we test whether an auxiliary objective can turn that organization into goal-relative ordering without discarding predictive state information. For a state latent z and goal latent g, define

$$\begin{aligned} d(z,g) &= \tfrac{1}{2}\left(1 - \frac{z^\top g}{\max(\|z\|_2,\epsilon)\max(\|g\|_2,\epsilon)}\right), \\ p(z,g) &= 1 - d(z,g). \end{aligned} \tag{2}$$

We clip numerical excursions so that $d, p \in [0,1]$. The scalar is goal relative: high p means that z is close to this visual goal, not that it has a task-independent completion percentage.

For each training sequence, we choose its goal h macro steps in the future and write $g = E_\theta(o_h)$. Let z_i and $\hat{z}_i$ denote the observed and imagined latents, respectively, with $\hat{z}_0 = z_0$. Define $d_i^O = d(z_i, g)$ and $d_i^I = d(\hat{z}_i, g)$, so $d_0^O = d_0^I = d_0$, where $d_0 = d(z_0, g)$. For r remaining steps and model-step discount $\gamma \in (0,1)$, $y_r = 1 - \gamma^r$ is a distance-to-go target; the corresponding progress target is $1 - y_r = \gamma^r$. We use a direct target-matching term, a Bellman term, and an imagined–observed consistency term:

$$\begin{aligned} \mathcal{L}_{\text{direct}} = \frac{1}{2h+2}\Big[&(d_0 - y_h)^2 + d(g,g)^2 \\ &+ \sum_{P\in\{I,O\}} \sum_{i=1}^{h} (d_i^P - y_{h-i})^2\Big], \end{aligned} \tag{3}$$

$$\mathcal{L}_B = \frac{1}{2h} \sum_{P\in\{I,O\}} \sum_{i=0}^{h-1} \left[d_i^P - (c + \gamma d_{i+1}^P)\right]^2 , \tag{4}$$

$$\mathcal{L}_{\text{cons}} = \frac{1}{h}\sum_{i=1}^{h}\left[d_i^I - d_i^O\right]^2 , \tag{5}$$

where $c = 1 - \gamma$. The endpoint term $d(g,g)^2$ explicitly anchors zero distance at the selected goal.

The progress loss is $\mathcal{L}_{\text{prog}} = \mathcal{L}_{\text{direct}} + \mathcal{L}_B + 0.25\mathcal{L}_{\text{cons}}$, and the full objective is

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + 0.09\mathcal{L}_{\text{SIGReg}} + \alpha_e \mathcal{L}_{\text{prog}}, \quad \alpha_e \le 0.002. \tag{6}$$

Here $\mathcal{L}_{\text{pred}}$ and $\mathcal{L}_{\text{SIGReg}}$ are the original LeWM prediction and regularization losses, and α_e is the epoch-dependent progress weight. Bellman targets remain in the computation graph rather than being detached. The goal and horizon construct the loss but are never predictor tokens. Because Equation 2 has no learned head, its gradients organize the shared encoder and predictor latent instead of training a separate progress network.

The auxiliary weight preserves predictive state information: a dominant scalar order could erase pose or contact, while an expert trajectory establishes only local order. We therefore keep SIGReg and next-latent prediction dominant and evaluate goal ordering jointly with prediction MSE; straightness alone is not evidence of progress.

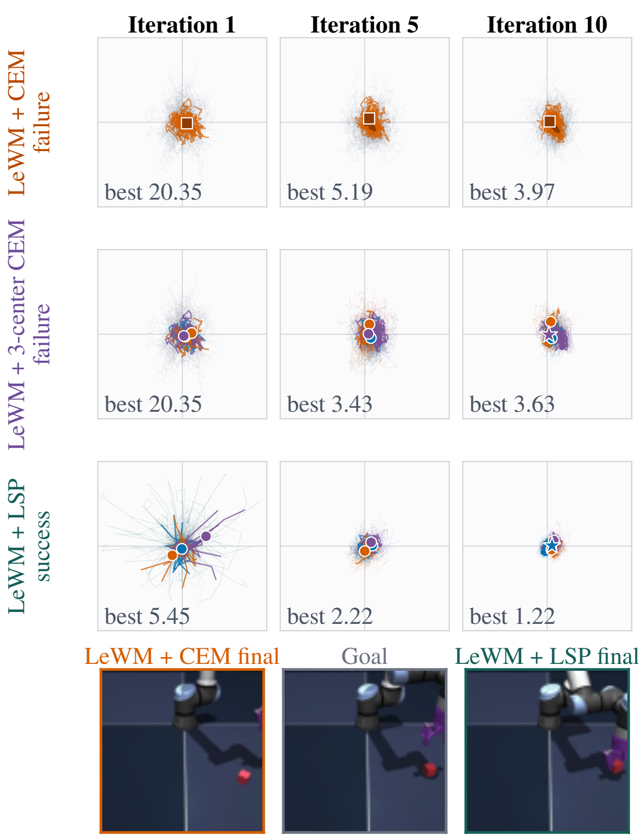

Figure 2: Illustrative LSP search evolution on a matched Cube replay. Progress-conditioned proposals expose a lower-cost mode in the first population and refinement preserves it to success; LeWM + CEM and LeWM + 3-center CEM fail. Values are terminal LeWM costs; lower is better.

LSP Search

LSP search turns the learned progress coordinate into a two-stage interface to sampling-based planning. It uses progress only to initialize the proposal population, then applies balanced multi-center refinement while retaining the terminal world-model cost as the sole evaluator.

Progress-Conditioned Proposal Initialization. After fine-tuning, we freeze the world model and pair offline action windows A with $p(E(o), E(o_g))$, using no reward, success, contact, or phase labels. The endpoint o_g lies a fixed nominal horizon after the start image o. These pairs encode local goal-relative compatibility; proposal diagnostics quantify the shift to online replanning.

A matched Cube replay makes the proposal-to-refinement interface concrete (Figure 2). LeWM + CEM and LeWM + 3-center CEM share the same zero-initialized first population, whereas LeWM + LSP replaces half of that population with progress-conditioned proposals, enters refinement from a lower-cost mode, and preserves it to success. The checkpoint, current–goal pair, solver seed, terminal cost, and rollout budget are identical across the three searches. Because the replay was selected after evaluation, it illustrates the mechanism rather than providing independent evidence; aggregate proposal and control tests appear in the Experiments section.

Before density fitting, we factor out known action symmetries in each applicable task. Cube provides a concrete example: actions are rotated by $-\phi(A)$, where $\phi(A) = \operatorname{atan2}(\sum_i a_i^y, \sum_i a_i^x)$ and a_i^x, a_i^y are the planar action components. This maps net planar displacement to $+x$ while preserving height, yaw, gripper state, and temporal order. Two-Room and Push-T use their corresponding planar transforms, whereas Reacher retains its native torque coordinates. Let A' denote the resulting sequence, with $A' = A$ for Reacher, and let $a = \operatorname{vec}(A')$. We standardize $\xi = [p; a]$, fit a K-component full-covariance mixture $q(p, a)$ with weights π_k, and transform its parameters back to raw progress/action coordinates. Partitioning component k into progress and action blocks gives

$$\begin{aligned}
\tilde{\rho}_k(p) &= \pi_k \mathcal{N}(p; \mu_k^p, \Sigma_k^{pp}), \\
\rho_k(p) &= \frac{\tilde{\rho}_k(p)}{\sum_{\ell=1}^K \tilde{\rho}_\ell(p)}, \\
\mu_k^{a|p} &= \mu_k^a + \Sigma_k^{ap}(\Sigma_k^{pp})^{-1}(p - \mu_k^p), \\
\Sigma_k^{a|p} &= \Sigma_k^{aa} - \Sigma_k^{ap}(\Sigma_k^{pp})^{-1}\Sigma_k^{pa},
\end{aligned} \tag{7}$$

$$q(a \mid p) = \sum_{k=1}^K \rho_k(p) \mathcal{N}(a; \mu_k^{a|p}, \Sigma_k^{a|p}). \tag{8}$$

The components are unsupervised action alternatives, not discrete phases. Conditioning the joint density retains both multiple components and within-component covariance at the same progress value; unlike direct regression, a query specifies neither one direction nor one control sequence. At runtime, samples a are unflattened and mapped back through the task transform (the identity for Reacher), inducing the raw action-sequence density $q(A \mid p)$; the terminal world-model cost then ranks their concrete directions and realizations.

Balanced Multi-Center Refinement. At each replan, LSP search queries $p(z_t, z_g)$ once and mixes samples from the base CEM distribution with $q(A \mid p)$. Assume N is even and that N and K_e are divisible by M. Equation 1 ranks the population and defines a global elite set $\mathcal{E}$ of size K_e. With action sequences understood as vectors, let D_0 be the diagonal matrix of initial action standard deviations. Rather than fit one Gaussian, LSP search seeks a disjoint partition $\biguplus_{m=1}^M \mathcal{E}_m = \mathcal{E}$, with $|\mathcal{E}_m| = K_e/M$, that minimizes

$$\min_{\{\mathcal{E}_m, \mu_m\}_{m=1}^M} \sum_{m=1}^M \sum_{A \in \mathcal{E}_m} \left\| D_0^{-1}(A - \mu_m) \right\|_2^2. \tag{9}$$

We approximate Equation 9 with farthest-first initialization and greedy equal-capacity assignment. Equal capacity prevents a dense group from consuming every elite slot; standardization prevents high-range action coordinates from dominating the split.

Each center then evolves an independent local CEM from its assigned elites while the total candidates and elites remain fixed. Initial center variances combine local residual

Method	Success (%) ↑ 2-Room	Reach.	Push-T	Cube
Reported in LeWM, Figure 6 (Maes et al. 2026)				
PLDM (Sobal et al. 2025)	97	78	78	65
DINO-WM (Zhou et al. 2025)	100	79	74	86
LeWM (Maes et al. 2026)	87	86	96	74
Our matched-budget reproduction and extensions				
LeWM + CEM	80.7	81.3	82.0	70.0
LeWM + LSP shaping + CEM	81.3	80.0	86.7	69.3
LeWM + LSP	94.0	93.3	86.7	84.7
Δ vs. control (pp)	**+13.3**	**+12.0**	**+4.7**	**+14.7**

Table 1: Closed-loop success. *Top:* results reported in LeWM Figure 6 (Maes et al. 2026). *Bottom:* our matched-budget reproduction and LSP extensions, averaged over three paired seeds. Complete seed-level results and paired 95% intervals are provided in the supplementary material.

Algorithm 1 One LSP search replan

Require: (z_t, z_g), cost C_θ, density $q(A \mid p)$, counts (N, K_e, M, J)

Set $p_t = p(z_t, z_g)$ with Eq. 2; sample $N/2$ base and $N/2$ proposal sequences.
Score all sequences with C_θ; retain the global top K_e.
Partition elites into M equal-capacity centers with Eq. 9.
for CEM iterations $j = 1, \ldots, J-1$ **do**
    Sample N/M sequences per center and score all with C_θ.
    Update each center from its local top K_e/M.
end for
Select the center with lowest mean local-elite cost; return its elite mean.

and global elite variance through shrinkage. After the last iteration, LSP search selects the center with the lowest mean local-elite cost and returns its elite mean. The returned mean is not re-scored, but every elite and center-selection statistic that produced it uses only Equation 1. Progress-conditioned sampling is restricted to the initial population.

The pseudocode makes the division of labor explicit (Algorithm 1). There is no rollout-level progress reranking, online reward, phase classifier, persistent proposal bank, or online model update. Bad initial proposals implicate the progress–action density; useful proposals that disappear implicate the optimizer. The learned dynamics and terminal goal cost retain responsibility for candidate and center selection.

Experiments

Our evaluation follows LSP shaping from latent geometry to the closed-loop behavior of LeWM + LSP. We first test whether LSP shaping induces goal-relative geometry in LeWM's shared latent without materially degrading next-latent prediction. We then measure end-to-end utility across four visual-control tasks using paired evaluations with matched world-model rollout budgets; same-checkpoint

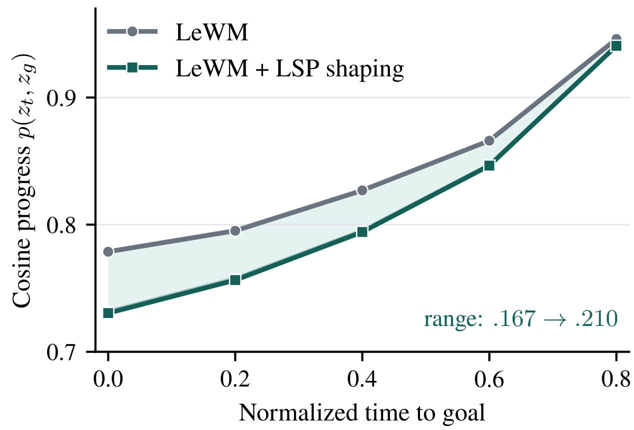

Figure 3: Goal-relative progress on 50 held-out Cube trajectories. LeWM + LSP shaping expands the nontrivial progress range from 0.167 to 0.210 by separating early states more strongly from the goal.

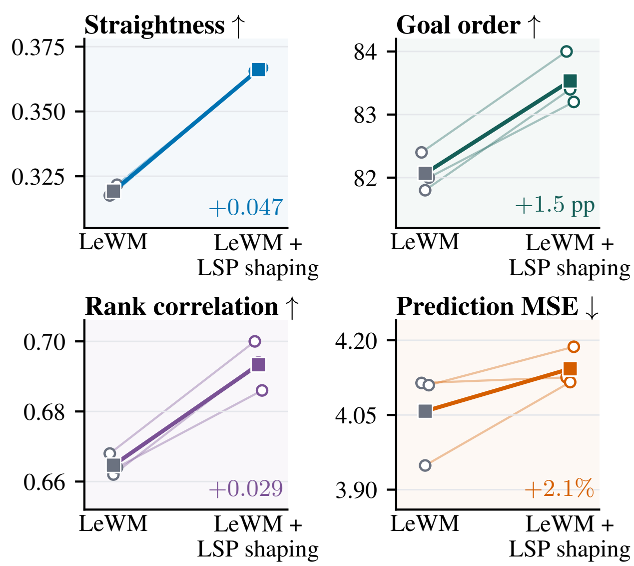

Figure 4: Paired latent-geometry diagnostics on the same trajectories. LeWM + LSP shaping improves straightness, goal order, and rank correlation, with a 2.1% increase in prediction MSE.

comparisons isolate the contribution of LSP search. Finally, Cube proposal diagnostics, component controls, and runtime measurements probe the sources of the gains and quantify their computational cost. Together, these experiments distinguish representational effects, planning-interface behavior, and closed-loop outcomes.

Experimental Setup and Implementation

We evaluate LSP with LeWM on Two-Room navigation, Reacher motion planning, Push-T pushing, and OGBench-Cube manipulation (Park et al. 2025; Maes et al. 2026). Three outcome-blind paired replicates each use 50 fixed current–

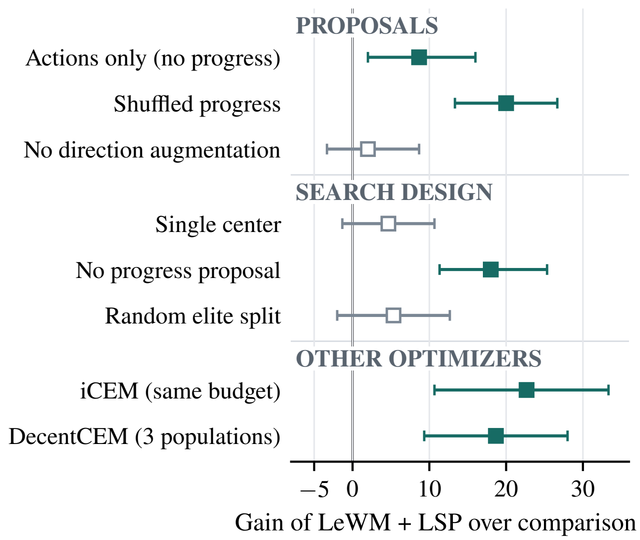

Figure 5: Cube controls under matched rollout budgets. Each row removes or replaces one LeWM + LSP design choice. Squares and whiskers show pooled paired gains and hierarchical 95% CIs; positive values favor LeWM + LSP, and filled squares exclude zero.

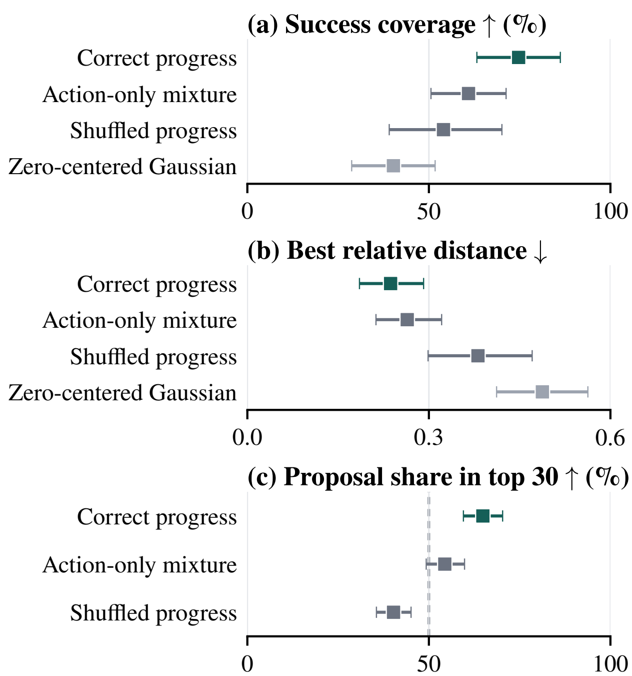

Figure 6: Cube first-population diagnostics. Squares are pooled means and whiskers are hierarchical 95% CIs. Higher is better in (a,c), while lower is better in (b). Panel (c) reports learned-proposal share against an equal-size zero-centered branch; the dashed line marks equal representation.

goal pairs with a 25-step goal offset and a 50-step execution budget, yielding 150 binary outcomes per condition. Within each task, all conditions share pair order, solver seed, success criterion, action coordinates, and candidate-rollout budget.

Starting from the official task-specific weights, we fine-tune the shared 192-dimensional backbone for three epochs, freeze it, and fit an eight-component full-covariance density to 5,000 held-out action windows. Every planner optimizes five macro actions with $N = 300$ candidates and $K_e = 30$ elites. LSP search uses $M = 3$ centers and splits its first population evenly between the original zero-centered Gaussian and progress-conditioned proposals. Two-Room, Reacher, and Cube use ten CEM iterations; Push-T uses 30. All planners therefore make the same number of candidate dynamics rollouts within a task.

LeWM + CEM receives the same fine-tuning budget without LSP shaping. *LeWM + LSP shaping + CEM* uses the shaped checkpoint with the same CEM planner, while *LeWM + LSP* adds LSP search to that checkpoint. We estimate paired percentage-point effects with 10,000 hierarchical bootstrap resamples, first over replicates and then over shared pair indices. On Cube, fixed-budget iCEM (Pinneri et al. 2021) and 3-population DecentCEM (Zhang et al. 2022) provide matched-rollout optimizer controls. The supplementary material consolidates full implementation details, seed-level outcomes, and additional controls.

Progress Geometry in Predictive Latents

The first question is whether LSP shaping gives the shared predictive latent a usable notion of position along a goal-directed trajectory. We compare matched checkpoints that start from the same released task weights and receive the same three-epoch optimization budget; only LeWM + LSP shaping includes the progress objective. On 50 held-out Cube trajectories, shaping separates early states more strongly from the goal while converging to the same endpoint. The nontrivial range $p(z_{0.8}, z_g) - p(z_0, z_g)$ expands from 0.167 to 0.210, with paired seed changes of 0.043, 0.044, and 0.042 (Figure 3). We exclude $t = 1$ because $p(z_g, z_g) = 1$ by construction. The consistent expansion across seeds shows that shaping increases the dynamic range of the goal-relative coordinate rather than merely changing one checkpoint.

We next ask whether the broader progress profile corresponds to more ordered temporal geometry without sacrificing predictive content. Across the same paired checkpoints, LSP shaping raises temporal straightness from 0.319 to 0.366, goal-ordering accuracy from 82.1% to 83.5%, and within-trajectory rank correlation from 0.665 to 0.693; next-latent MSE changes from 0.00406 to 0.00414 (Figure 4). Thus the representation becomes more ordered with only a 2.1% increase in prediction error.

Together, the profile and geometry diagnostics establish a consistent but modest representation change: LSP shaping exposes progress while largely preserving next-latent prediction. This result supports the representation claim, but does not by itself establish that shaping is necessary for control success.

Closed-Loop Actionability

The representation is useful only if it improves decisions made under the same world-model rollout budget. Relative to LeWM + CEM, LeWM + LSP gains 13.3 points on Two-Room (95% CI $[6.0, 21.3]$), 12.0 on Reacher ($[3.3, 20.7]$), 4.7 on Push-T ($[-2.0, 11.3]$), and 14.7 on Cube ($[8.0, 21.3]$). The gains are supported on Two-Room, Reacher, and Cube, whereas Push-T remains unresolved because its interval crosses zero. Table 1 reports the corresponding success rates and includes values from LeWM Figure 6 only as descriptive context, not for cross-source significance claims.

To separate search from checkpoint quality, we compare LeWM + LSP with LeWM + LSP shaping + CEM on the same shaped checkpoint. Adding LSP search changes success by $+12.7$ points on Two-Room ($[6.0, 20.7]$), $+13.3$ on Reacher ($[5.3, 21.3]$), 0.0 on Push-T ($[-6.0, 6.0]$), and $+15.3$ on Cube ($[6.7, 23.3]$). The same three tasks improve while Push-T ties, showing that the resolved closed-loop gains extend beyond representation shaping alone.

What Drives the Gain?

The end-to-end gain could come from correct progress conditioning, the search structure, or simply a stronger sampling optimizer. We isolate these explanations on Cube under the same rollout budget. Replacing correct progress with an action-only mixture reduces success by 8.7 points ($[2.0, 16.0]$), while shuffling progress reduces it by 20.0 points ($[13.3, 26.7]$). These are the clearest resolved component effects in Figure 5, identifying correct task-stage conditioning as the best-supported source of the Cube gain.

The remaining controls narrow the interpretation. Shaping alone changes Cube success by -0.7 points ($[-5.3, 4.0]$), and the effects of direction augmentation, using 3 rather than 1 center, and structured rather than random elite splitting all cross zero. In contrast, LeWM + LSP exceeds 3 centers without progress proposals by 18.0 points, fixed-budget iCEM by 22.7, and 3-population DecentCEM by 18.7, with all 3 intervals excluding zero. Generic multi-center search therefore does not reproduce the gain; the evidence more strongly supports the progress-conditioned proposal interface than every individual refinement choice.

Does Progress Improve Proposals?

If correct progress drives the control gain, its effect should be visible before CEM performs any update. On initially unsolved Cube pairs, correct progress raises first-population coverage to 74.7%, compared with 40.2% for the zero-centered Gaussian, 54.0% for shuffled progress, and 60.9% for the action-only mixture. It also lowers the best relative goal distance to 0.237, from 0.488, 0.381, and 0.264, respectively. The simulator coverage and frozen-model distance therefore agree: correct progress places useful actions in the initial population more often and with better goal proximity (Figure 6). This pre-update effect connects the learned coordinate to planning without relying on progress as a rollout score.

Planning-Time Cost

Matching world-model rollout counts controls the dominant model-evaluation budget, but LSP still adds proposal sampling and balanced center updates. Synchronized profiling measures a 1.7–4.8% mean planning-time overhead across tasks; the largest upper confidence bound is 6.0% on Cube (Table 2).

Task	CEM	LSP	Overhead [95% CI]
Two-Room	333.72	344.93	3.4 $[3.1, 3.6]$
Reacher	334.97	343.18	2.5 $[2.1, 2.8]$
Push-T	1002.23	1019.37	1.7 $[1.1, 2.3]$
Cube	339.31	355.59	4.8 $[3.7, 6.0]$

Table 2: Planning time in ms/replan under matched world-model evaluation counts. CEM and LSP use the same checkpoint within each task; overhead is relative to CEM.

Conclusion

We begin from evidence that predictive latents can develop temporal organization and ask how that structure can inform planning. LSP shaping turns that structure into a cosine progress query; LSP search uses it once to initialize stage-compatible proposals and then refines balanced elite modes, while terminal latent MSE remains responsible for ranking candidates and centers. Under paired fixed-rollout evaluation, LeWM + LSP improves Two-Room, Reacher, and Cube over LeWM + CEM and LeWM + LSP shaping + CEM, while maintaining comparable Push-T performance. Cube controls identify correct progress conditioning as the best-supported source of stronger initial proposals, while shaping and multi-center refinement remain individually unresolved. Together, these results show that the temporal organization observed in LeWM can become an actionable progress coordinate without displacing learned dynamics.

Limitations

These results establish LSP within LeWM across four visual-control tasks while leaving several natural extensions open. Future work can test other latent architectures and out-of-distribution goals, relax the fixed-offset action-window and known-symmetry assumptions, and further reduce the measured 1.7–4.8% planning-time overhead.

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